CEFET/RJ

DAL & Comunidade

ESTRUTURA DE REPETIÇÃO

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Exemplo

RStudio

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exemplo.R*

```r
weight <- 60
height = 1.75
subject <- "A"
healthy <- TRUE
bmi <- weight/height^2  #Body mass index (BMI)
bmi
```

Console

```
R 4.2.1 · C:/R/main/
> weight <- 60
> height = 1.75
> subject <- "A"
> healthy <- TRUE
> bmi <- weight/height^2  #Body mass index (BMI)
> bmi
[1] 19.59184
>
```

Environment

Values	
bmi	19.5918367346939
healthy	TRUE
height	1.75
subject	"A"
weight	60

Files: C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Exemplo

RStudio

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exemplo.R*

```r
weight <- c(60, 72, 57, 90, 95, 72)
height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
subject <- c("A", "B", "C", "D", "E", "F")
```

Console

```
> weight <- c(60, 72, 57, 90, 95, 72)
> height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
> subject <- c("A", "B", "C", "D", "E", "F")
>
```

Environment

Values	
bmi	19.5918367346939
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
subject	chr [1:6] "A" "B" "C" "D" "E"...
weight	num [1:6] 60 72 57 90 95 72

Files

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Estrutura de repetição for

RStudio

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exemplo.R*

```r
bmi <- 0
for (i in 1:length(weight)) {
  bmi[i] <- weight[i]/height[i]^2
}
bmi
```

Console

```
> bmi <- 0
> for (i in 1:length(weight)) {
+   bmi[i] <- weight[i]/height[i]^2
+ }
> bmi
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
>
```

Environment

bmi	num [1:6] 19.6 22.2 20.9 24.9...
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	6L
subject	chr [1:6] "A" "B" "C" "D" "E"...

Files

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Inspecionando os cálculos

RStudio

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exemplo.R*

```r
bmi <- 0
for (i in 1:length(weight)) {
  bmi[i] <- weight[i]/height[i]^2
  print(bmi)
}
bmi
```

Console — R 4.2.1 · C:/R/main/

```
> bmi <- 0
> for (i in 1:length(weight)) {
+   bmi[i] <- weight[i]/height[i]^2
+   print(bmi)
+ }
[1] 19.59184
[1] 19.59184 22.22222
[1] 19.59184 22.22222 20.93664
[1] 19.59184 22.22222 20.93664 24.93075
[1] 19.59184 22.22222 20.93664 24.93075 31.37799
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
> bmi
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
>
```

Environment — Global Environment

values	
bmi	num [1:6] 19.6 22.2 20.9 24.9...
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	6L
subject	chr [1:6] "A" "B" "C" "D" "E"...

Files — C: > R > main

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Depurando estrutura de repetição for

RStudio

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exemplo.R

```
1  bmi <- 0
2  for (i in 1:length(weight)) {
3    bmi[i] <- weight[i]/height[i]^2
4  }
5  bmi
6
```

3:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> debugSource("C:/Users/eduar/OneDrive/Aulas/R/exemplo.R")
called from: eval(expr, p)
Browse[1]> n
debug at C:/Users/eduar/OneDrive/Aulas/R/exemplo.R#3: bmi[i] <- wei
ght[i]/height[i]^2
Browse[2]>
```

Environment History Connections Tutorial

172 MiB

Global Environment

Values

bmi	0

Traceback

```
[Debug source] at exemplo.R:3
```

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Criando a função

Clique na vassoura para limpar as variáveis

Removendo varáveis

RStudio

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exemplo.R*

```
rm(bmi)
exists("bmi")
```

Console

```
> rm(bmi)
> exists("bmi")
[1] FALSE
>
```

Environment

healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	1L
subject	chr [1:6] "A" "B" "C" "D" "E"...
weight	num [1:6] 60 72 57 90 95 72

Files

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Estrutura de repetição while

RStudio

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exemplo.R*

```r
i <- 1
bmi <- 0
while (i <= length(weight)) {
  bmi[i] <- weight[i]/height[i]^2
  i <- i + 1
}
```

7:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> i <- 1
> bmi <- 0
> while (i <= length(weight)) {
+   bmi[i] <- weight[i]/height[i]^2
+   i <- i + 1
+ }
>
```

Environment History Connections Tutorial

Global Environment

Values

bmi	num [1:6] 19.6 22.2 20.9 24.9...
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	7
subject	chr [1:6] "A" "B" "C" "D" "E"...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Inspecionando estrutura de repetição while

RStudio

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exemplo.R*

```r
i <- 1
bmi <- 0
while (i <= length(weight)) {
  bmi[i] <- weight[i]/height[i]^2
  print(bmi)
  i <- i + 1
  print(i)
}
```

9:1 (Top Level) R Script

Console

R 4.2.1 · C:/R/main/

```
[1] 19.59184
[1] 2
[1] 19.59184 22.22222
[1] 3
[1] 19.59184 22.22222 20.93664
[1] 4
[1] 19.59184 22.22222 20.93664 24.93075
[1] 5
[1] 19.59184 22.22222 20.93664 24.93075 31.37799
[1] 6
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
[1] 7
```

Environment

Values

bmi	num [1:6] 19.6 22.2 20.9 24.9...
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	7
subject	chr [1:6] "A" "B" "C" "D" "E"...

Files

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Depurando estrutura de repetição while

RStudio

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exemplo.R

```
1  i <- 1
2  bmi <- 0
3  while (i <= length(weight)) {
4    bmi[i] <- weight[i]/height[i]^2
5    i <- i + 1
6  }
7
```

5:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

Next Continue Stop

```
[1] 5
[1] 19.59184 22.22222 20.93664 24.93075 31.37799
[1] 6
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
[1] 7
> debugSource("C:/Users/eduar/OneDrive/Aulas/R/exemplo.R")
Called from: eval(expr, p)
Browse[1]> n
debug at C:/Users/eduar/OneDrive/Aulas/R/exemplo.R#5: i <- i + 1
Browse[2]>
```

Environment History Connections Tutorial

171 MiB List

R Global Environment

Values

bmi	19.5918367346939
healthy	TRUE
height	num [1:6] 1.75 1.8 1.65 1.9 1...
i	1
subject	chr [1:6] "A" "B" "C" "D" "E"...

Traceback Show internals

[Debug source] at exemplo.R:5

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2

Encapsulando cálculo do bmi como função

RStudio

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exemplo.R

```r
compute_bmi <- function(weight, height) {
  i <- 1
  bmi <- 0
  while (i <= length(weight)) {
    bmi[i] <- weight[i]/height[i]^2
    i <- i + 1
  }
  return(bmi)
}

bmi <- compute_bmi(weight, height)
```

11:34 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/exemplo.R")
>
```

Environment History Connections Tutorial

Global Environment

Values	
bmi	num [1:6] 19.6 22.2 20.9 24.9...
height	num [1:6] 1.75 1.8 1.65 1.9 1...
subject	chr [1:6] "A" "B" "C" "D" "E"...
weight	num [1:6] 60 72 57 90 95 72
Functions	
compute_bmi	function (weight, height)

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2

Implementando a função do jeito certo

RStudio

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exemplo.R

```r
compute_bmi <- function(weight, height) {
  resposta <- weight/height^2
  return(resposta)
}

bmi <- compute_bmi(weight, height)
bmi
```

Console — R 4.2.1 · C:/R/main/

```
>
```

Environment — Global Environment

Values	
bmi	num [1:6] 19.6 22.2 20.9 24.9…
height	num [1:6] 1.75 1.8 1.65 1.9 1…
subject	chr [1:6] "A" "B" "C" "D" "E"…
weight	num [1:6] 60 72 57 90 95 72
Functions	
compute_bmi	function (weight, height)

Files — C: > R > main

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2

Depurando a função

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R

```
1  compute_bmi <- function(weight, height) {
2    resposta <- weight/height^2
3    return(resposta)
4  }
5
6  bmi <- compute_bmi(weight, height)
7  bmi
8
```

8:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/exemplo.R")
> debugSource("C:/Users/eduar/OneDrive/Aulas/R/exemplo.R")
Called from: eval(expr, p)
Browse[1]> n
debug at C:/Users/eduar/OneDrive/Aulas/R/exemplo.R#3: return(resp
osta)
Browse[2]>
```

Environment History Connections Tutorial

173 MiB

R compute_bmi()

Values

height	num [1:6] 1.75 1.8 1.65 1.9 1...
resposta	num [1:6] 19.6 22.2 20.9 24.9...
weight	num [1:6] 60 72 57 90 95 72

Traceback Show internals

```
compute_bmi(weight, height) at exemplo.R:3
[Debug source] at exemplo.R:6
```

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2

Uso da função (escalares/vetores)

RStudio

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exemplo.R*

```
1 compute_bmi(80, 1.79)
2 compute_bmi(weight, height)
3
```

3:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> compute_bmi(80, 1.79)
[1] 24.96801
> compute_bmi(weight, height)
[1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
>
```

Environment History Connections Tutorial

Values

bmi	num [1:6] 19.6 22.2 20.9 24.9...
height	num [1:6] 1.75 1.8 1.65 1.9 1...
subject	chr [1:6] "A" "B" "C" "D" "E"...
weight	num [1:6] 60 72 57 90 95 72

Functions

compute_bmi	function (weight, height)

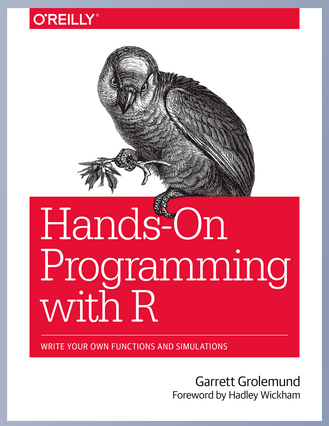
Files Plots Packages Help Viewer Presentation

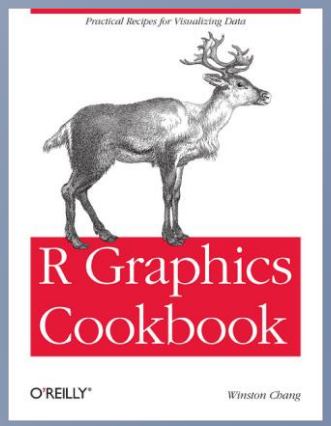
C: > R > main

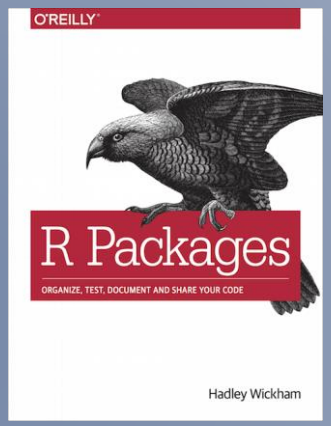
Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2

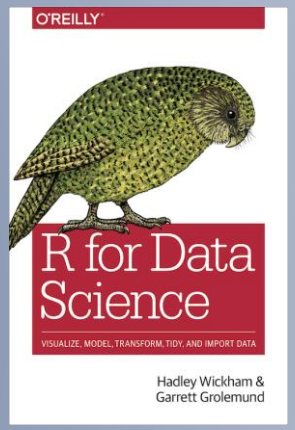
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/06-Estrutura-Repeticao.ipynb







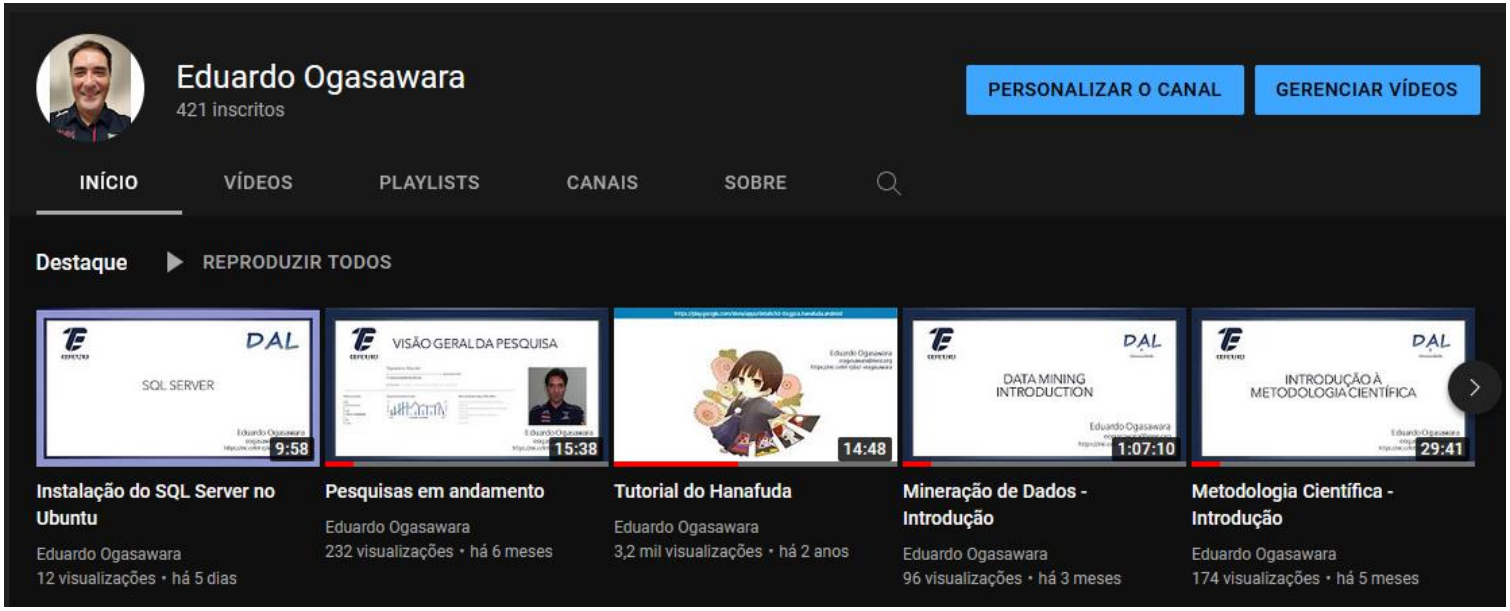


Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R